



Department of Electronic & Telecommunication Engineering

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PROJECT REPORT

Cheap Soldering Iron with Thyristor based Power Control

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Abstract

This report describes the design process and rationale behind designing a cheap thyristor-based power controlled soldering iron. A Potentiometer will adjust the firing angle of the thyristors. As A/C power will be used, there is no need for a commutating circuit to turn off the thyristor as the current passes through zero each cycle. The chosen firing angle will affect the rms of the power output to the heating element.

Acknowledgements.

We would like to thank laboratory staff of the Analog lab, Mr. Thilina Sameera Ambagaswatta, and many of our colleagues for their constant guidance and support.

Introduction

Solder irons are hand tools used for installation, repairs, and limited production work in electronics assembly. These irons function by passing an electric current through a resistive heating element which melts the solder via its metal tip.

However, these simple solder irons have many limitations which can lead to defective soldering.

This solution attempts to provide a solder iron where it is possible to control the power it is being supplied with.

This project is made for providing hobbyists with a cheap, power-controlled solder iron to carry out their interests without price being a concern.

Background

Solder tips are composed of copper cores plated with iron, which are susceptible to rust. As the temperature increases, it increases the rate of oxidation thus furthering rusting which can damage the tip. Thus, it required to keep the tip at an average lower temperature for tip preservation. It can be done by limiting the power supplied. This will reduce the tip damage and the need for constant tip cleaning, which can wear it out. If the tip is damaged, there is a potential for the copper to leak out. Rusty tips make for poor thermal conductivity which makes soldering hard.

In uncontrolled Solder Irons, the temperature is dependent on thermal equilibrium. When not in use, it will heat up till the generated heat is equal to the heat drained to the environment. Thus, when placing the tip on the solder, it can

burn and make the soldering bad.

Regular solder irons provide the same power to all soldering needs; thus, it is important adjust the power being delivered depending on the item being soldered. With power control, it is possible to set the power to match the kind of solder being used (hotter for lead-free), turn it down to solder delicate or heat-sensitive parts such as (ICs, CMOS.MOSFETs which are sensitive to small temperature excursions), or turn it up to work on large or heat-conductive terminals. Generally, for soldering small resistors and ICs, 15 Watts will suffice, however it is important to wait a bit in between joints for the tip to recover. For soldering larger things like 10-gauge copper wire, motor casings, or large heat sinks, extremely high power be may needed upwards of a 50-watt iron or more.

Low-voltage soldering irons (e.g. 12V) generally form part of a soldering station and are designed to be used with a temperature controller. A proper temperature-controlled soldering iron or station is extremely expensive. Thus, it is important to design a cheaper alternative with a similar degree of control.

Problem Statement

- Most solder iron tips are susceptible to rust and damage due to no power control
- Solder iron has the potential to burn PCBs due to heating up to excessive temperatures and there is a need for power control
- Differing power requirements for soldering different items are usually not met with conventional soldering irons

- Temperature controlled soldering irons are extremely expensive, so most hobbyists cannot afford them.

Requirements

- Use a thyristor-based power control system.
- It should be cheap
- Highly accurate and reliable power control
- High safety
- Increased lifetime

Alternative Approaches

Arduino-Based with slave opto-triac with master triac circuit.

In this implementation an Arduino controls the firing angle by varying the phase angle of a steady pulse wave which is fed into the opto-triac. Due to the presence of the Arduino this circuit can host additional elements such as a Thermocouple with a MAX6625 cold junction circuit, which can provide thermal feedback to improve accuracy. An OLED screen can also be used to view the present temperature.

However, it requires many DC power inputs for its Arduino microcontroller circuit and op amps (for zero-crossing detection). This increases its size. Additionally, it is the most expensive implementation due to the high cost of many of its components. There is a high resource waste as the microcontroller is for general purpose usage. It also requires lot of tuning for accurate temperature detection. It also requires special modifications done to the solder iron to include the thermocouple. Thus, it is extremely expensive.

BUCK converter Approach

This is a viable option, but this violates the main objective of the project of using a thyristor element.

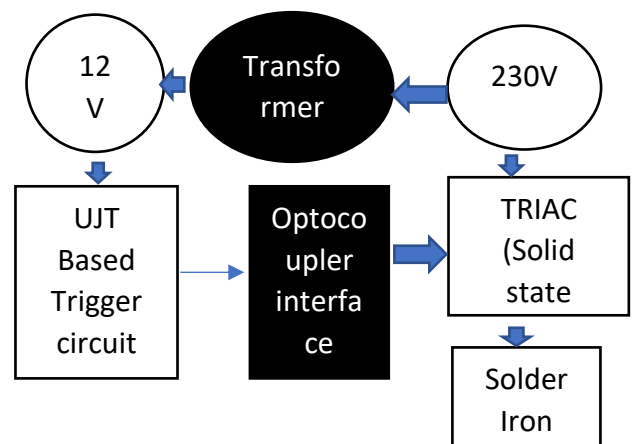
DIAC-TRIAC Approach

This is the cheapest approach, however there is increased risk of electrocution for to the user. Additionally, it is extremely unreliable and not accurate. Here the values of the resistors, capacitors need to be precise for the given voltage waveform. Since DIACs have extremely high breakover voltages it is nearly impossible to test the circuit via regular lab equipment.

Specification

The project uses TRIAC and an opto-TRIAC for controlling the power. This is AC Solid State Relay Circuit

It is triggered by a separate UJT based trigger circuit thus it is highly accurate. Since the trigger circuit is electrically isolated via an opto-TRIAC it is extremely reliable. With the inductive and optical coupling, it doesn't expose the use to the main power supply thus it is extremely safe. The power control means the tips of the soldering iron prevented from degrading.



Input Voltage	230V AC at 60Hz
Max Input Voltage	230V (peak)
Max Input Current	1 A
Load	2 Ohms

Design

This system consists of two modules. The first module is used to control the trigger signal which varies according a potentiometer. The second module controls the power supplied to the solder based on the trigger signal.

The first module consists of mainly quarter watt resistors, capacitors, bridge rectifier and a UJT device is powered from the secondary side of a transformer which reduces the voltage to 12V.

The second module consists of power triac, two-watt power resistors, and an opto-coupled TRIAC which interfaces the two circuits.

Features

Accuracy

Compared to other circuits, it is possible to accurate control the firing angle of the trigger signal by the potentiometer on the first module. Here the potentiometer controls the charging time of the capacitor. Best part of the circuit is that, there is no need for a zero-crossing detector. Additionally, firing angle is directly proportional to the potentiometer value, unlike the TRIAC /DIAC implementation.

Safety

This system consists of two separate circuits which are optically and inductively coupled from each other. Transformer also provide galvanic “electrical isolation” and DC isolation between the higher voltage on the primary side and the lower voltage on the secondary side, increasing the safety.

This increases the safety of the circuit as the user does not handle components connected to the main supply. It also provides complete isolation from any noise or voltage spikes present on the AC power supply line as well as zero-crossing detection of the sinusoidal waveform which reduces switching and inrush currents protecting any power semiconductors used from thermal stress and shock.

Cheapness

Other than the transformer, most of the components are cheap.

Flexibility

The phase control is implemented by the first module which is independent from the second module which handles the main power supply. The circuit is not affected by

the soldering iron impedance (which can change due temperature changes) Additionally it only controls the part of the AC sine wave delivered, and it doesn't depend on the its values unlike the TRIAC-DIAC circuit.

Implementation

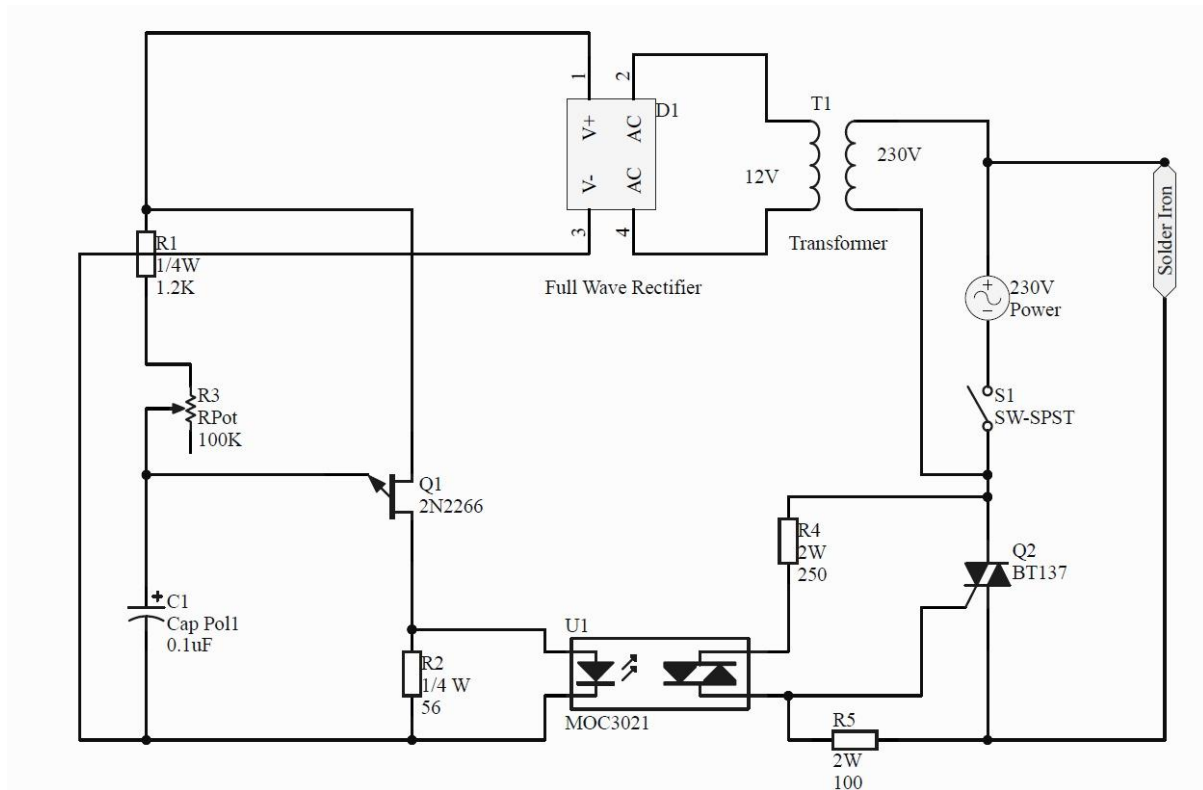


Figure 1: Schematic

UJT Trigger circuit

Generally, the equation of the time period of the UJT pulse is given by the following equation, where η is the intrinsic standoff ratio for a given UJT which varies between 0.8 and 0.4

$$T = R_3 C \ln \frac{1}{1 - \eta}$$

It is required to restrict the variation of the frequency of the pulse between 0s and (1/230) s for the device have a large degree of sensitivity.

For safety reasons R1 was selected as 1.2K and based on the calculations R3 was picked as 100K. A large capacitor value was selected to ensure the R3 was below 200K.

Solid state relay circuit

100 ohm resistor (R5) shunts any di/dt (transient) currents when the triac is OFF

eliminating false triggering. It also ties the gate terminal to MT1 ensuring the triac turns-off fully.

Consider that the output triac of the opto-isolator has switched ON at the peak value (90o) of the 230V (pk-pk) AC supply voltage. If the opto-triacs maximum current (that is the non-repetitive surge peak on state current) is 1 ampere peak, then the minimum value of series resistance require is $230/1 = 230\Omega$'s to the nearest preferred value. This value of 230 Ω 's will protect the opto-coupler output triac, as well as the gate of the power triac on a AC supply.

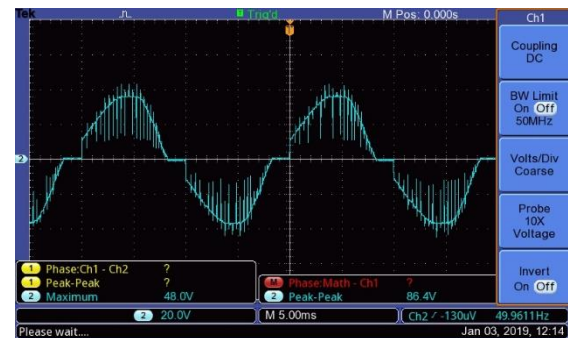
If the triac of the opto-isolator switches ON at the zero-crossover value (0o) of the 120VRMS AC supply voltage, then the minimum voltage required to supply the required 50mA gate drive current forcing

the switching triac into conduction will be:
 $180\Omega \times 50\text{mA} = 9.0\text{ volts}$. Then the triac
 fires into conduction when the sinusoidal
 Gate-to-MT1 voltage is greater than 9 volt

Experimental Results

The output was obtained for a 20V peak to
 peak supply to gather simulation results as
 high voltages cannot be read by an
 oscilloscope. The spikes can be eliminated
 using a varistor. Additionally, the light bulb
 was used to test the result.

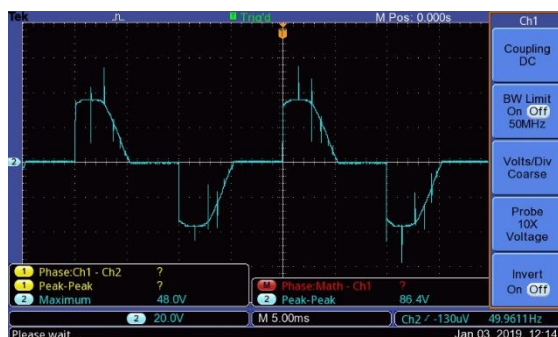
The result showed highly accurate and
 reliable power control.



The project was relatively cheap compared
 to the past projects and the net total spent
 on all the equipment for complete design
 was 600 /= Rupees. Thus, it fulfils our cost
 requirement.

To test safety, a mustimeter was used to
 check all the components of the circuit
 when it was plugged to the main power
 supply. All components which are exposed
 to the user registered below 13V which is
 safe for human contact. An enclosure was
 designed, however due to the time
 constraints on the project it wasn't
 possible to make it and fit it. (Refer
 appendix) It fulfilled the safety
 requirement.

It wasn't possible to experimentally verify
 the increased lifetime due to time
 constraints placed on the project.



Future Work

It is possible to include following components

- Varistors: can be used to remove the transient spikes on the output waveform.
- Snubber circuits: can be used to protect TRIAC and other switching devices.

Additionally, a thermal feedback via Analog means would increase the reliability of the design. This can be done via Opamps and a thermocouple.

Conclusions

It was possible to design a thyristor-based power control system with TRIACs and Optotriacs which is affordable to a hobbyist. The system was highly accurate, safe and displayed reliable power control.

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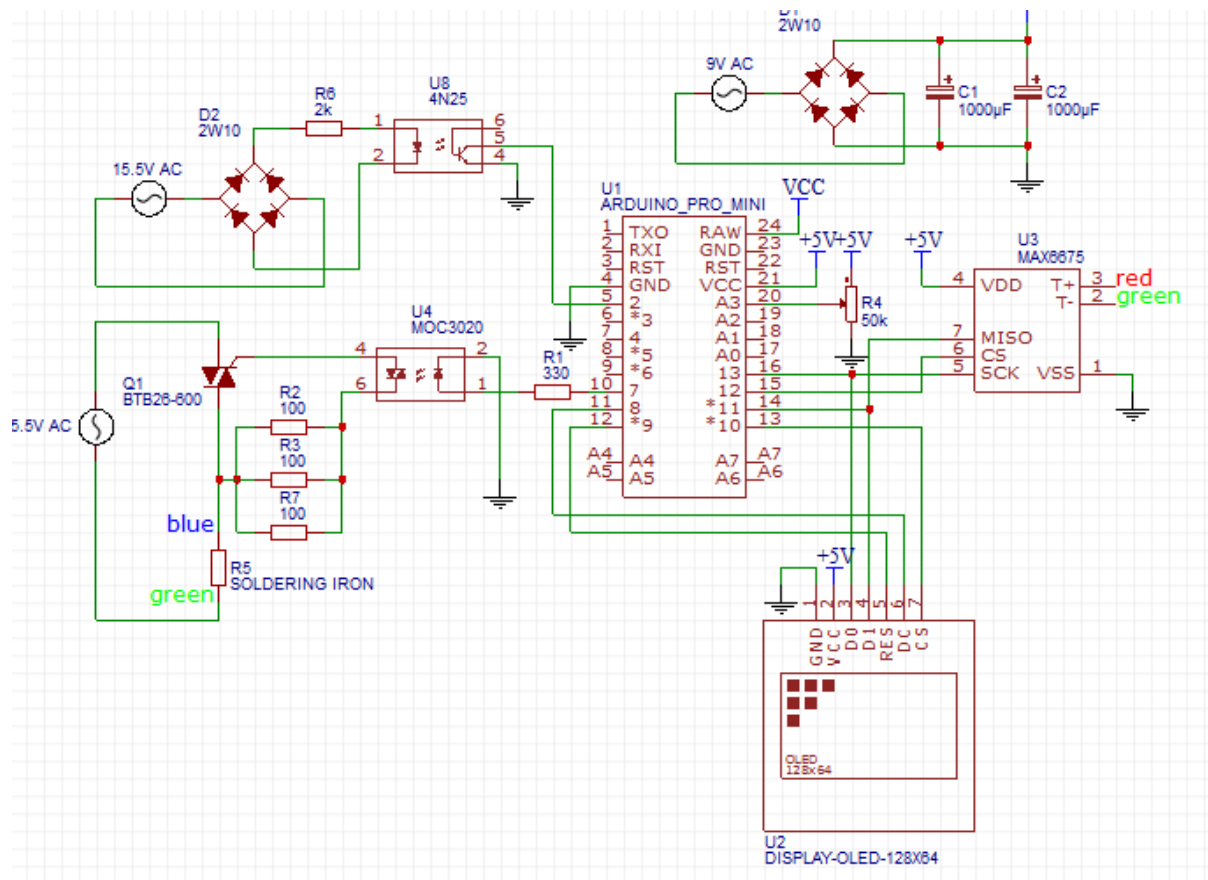
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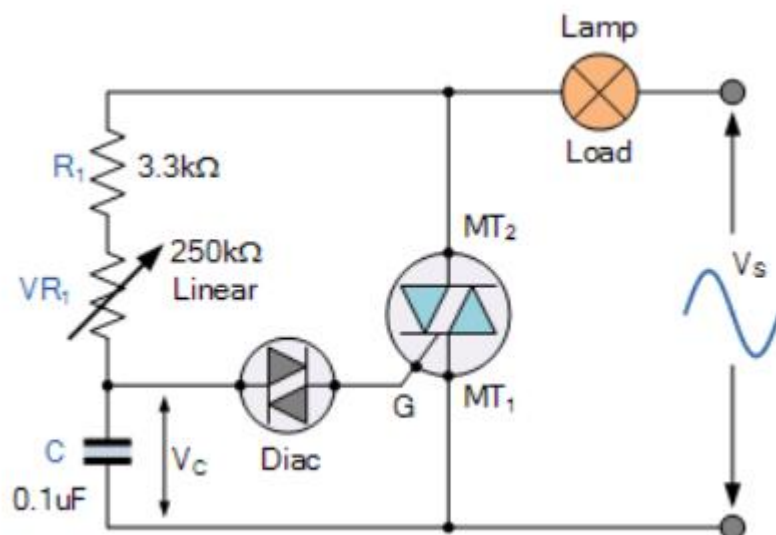
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Appendix

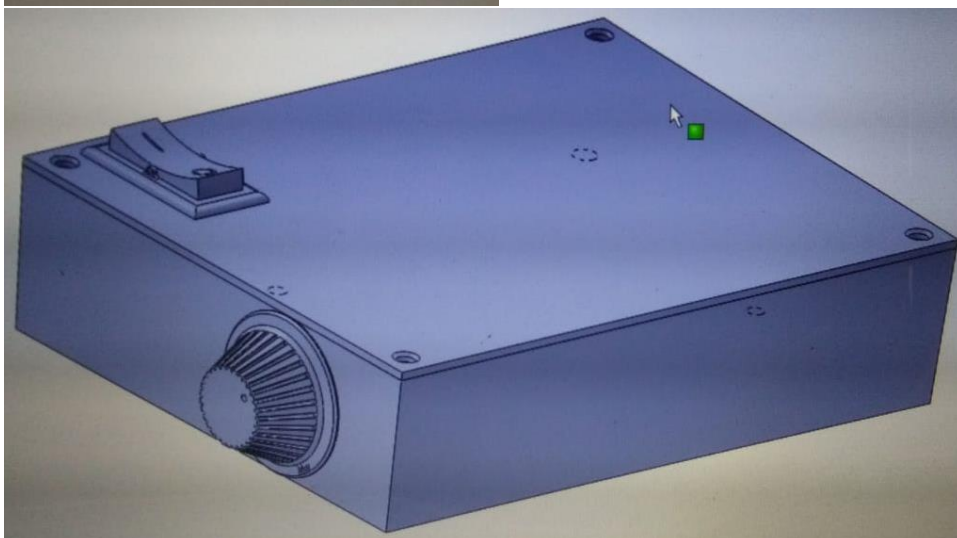
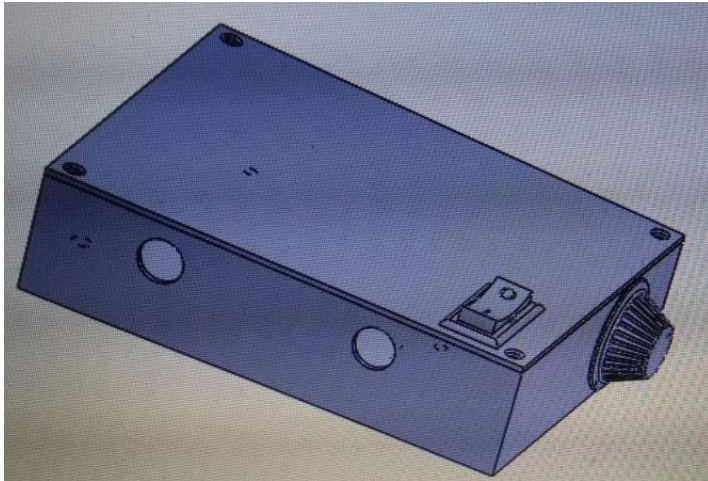
Arduino based version



Diac-Triac based version



Enclosure design for the circuit



PCB

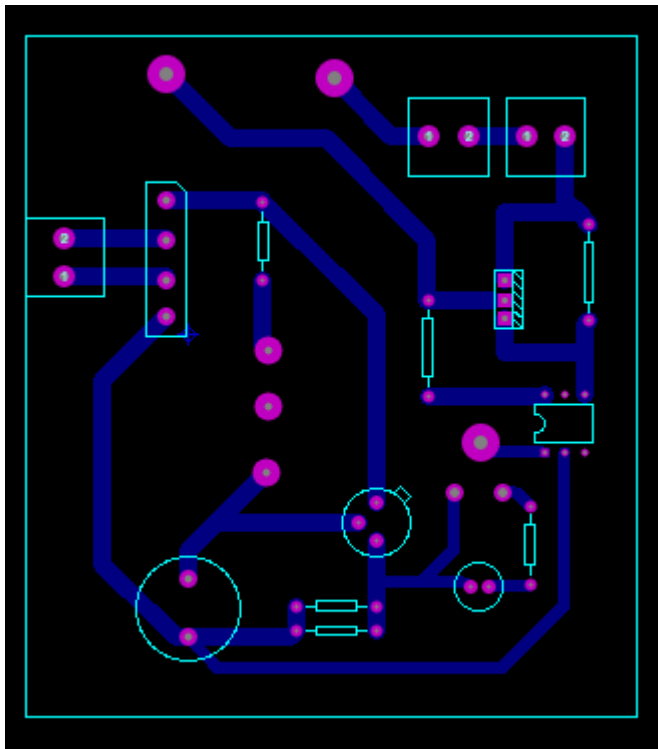


Figure 2: PCB Layout

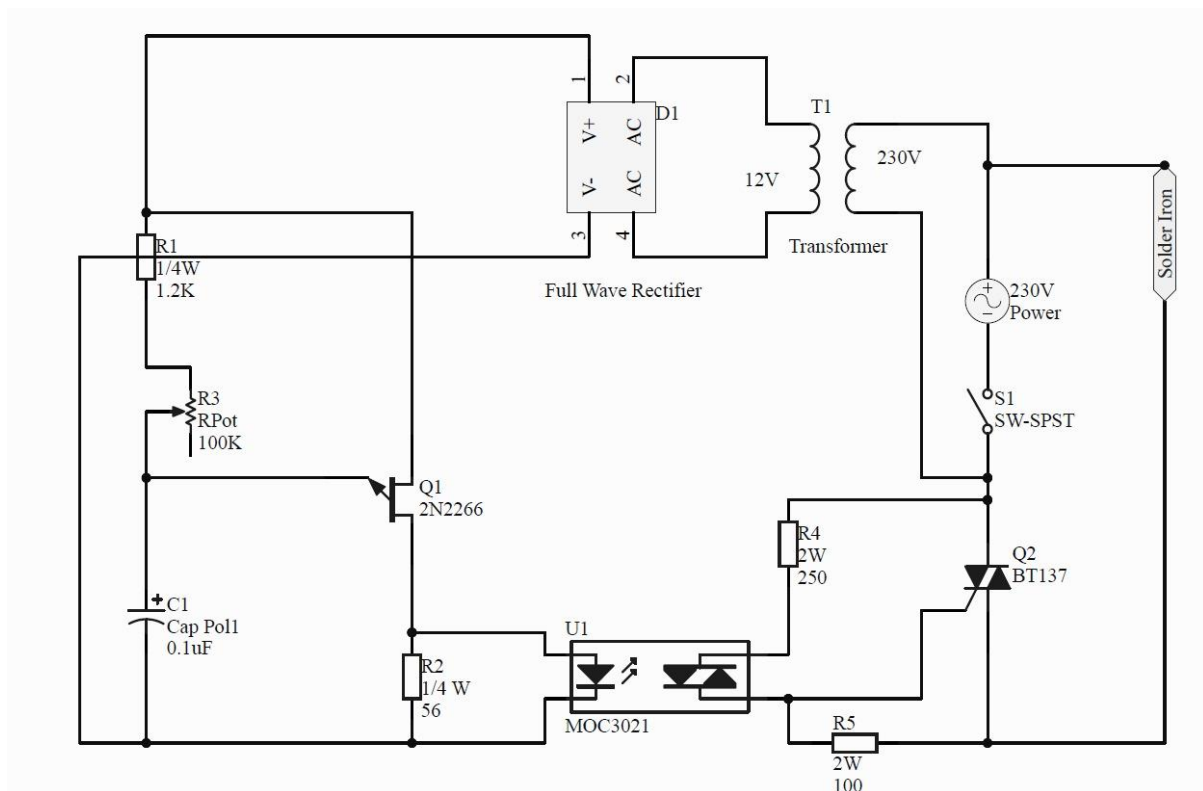


Figure 3: Schematic of the Circuit